

Erratum and Revisions of The Hash Function MOFANG

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Abstract

This document provides errata and revisions to the specification of MOFANG, a hash function family submitted to the NGCC program. The errata correct inaccuracies and ambiguities in the submitted specification and the revisions modify the key expansion to remove a structural symmetry between block-derived round keys. A complete list of changes is given in the table below. Accordingly, the reference and optimized implementations have been updated, and all known-answer test (KAT) vectors have been regenerated for all variants of the family.

Table 1: Summary of changes to the submitted specification.

#	Location	Change
1	Section 2.4.4	In Algorithm 3, in line 13, $B[0] \leftarrow \text{SBox}(A[0])$ is replaced by $B[0] \leftarrow \text{SBox}(A[1])$.
2	Section 2.4.4	In Algorithm 4, in line 18, $B[0] \leftarrow \text{SBox}(A[0])$ is replaced by $B[0] \leftarrow \text{SBox}(A[1])$.
3	Section 2.4.4	In Algorithm 4, in line 23, $B[x, y] \leftarrow A[y, 2x + y]$ is replaced by $C[x, y] \leftarrow B[y, 2x + y]$.

1 Revised Key Expansion

In the key expansion of the originally submitted specification, the S-box acts on the same row for every block, namely, the first of the three rows. In the revised version, the S-box is applied to the first and second rows, respectively, for the different blocks, in order to break the structural similarity between the two block-derived halves, which previously enabled identical round-key components across blocks. All other components remain unchanged. The complete Algorithms 3 and 4 are as follows. The typographical error in line 23 of Algorithm 4 has also been corrected to be consistent with the reference implementation.

Algorithm 3: The Key Expansion For MOFANG-256/512

Input: State M_1, M_2 (input length: $576 * 2$)

Output: State K

```
1  $A \leftarrow M_1$ 
  /* S-box for the first row for the first block */
2  $B[0] \leftarrow \text{SBox}(A[0])$ 
3 for  $x = 1$  to 2 do
4    $B[x] \leftarrow A[x]$ 
  /* Transposition of words for the first block */
5 for  $x = 0$  to 2 do
6   for  $y = 0$  to 2 do
7      $C[x, y] \leftarrow B[y, 2x + y]$ 
  /* ShiftRow for the first block */
8 for  $x = 0$  to 2 do
9   for  $y = 0$  to 2 do
10     $D[x, y] \leftarrow \text{RotR}(C[x, y], \rho_1(x, y))$ 
11  $K \leftarrow D$ 
12  $A \leftarrow M_2$ 
  /* S-box for the second row for the second block */
13  $B[0] \leftarrow \text{SBox}(A[1])$ 
14 for  $x = 1$  to 2 do
15    $B[x] \leftarrow A[x]$ 
  /* Transposition of words for the second block */
16 for  $x = 0$  to 2 do
17   for  $y = 0$  to 2 do
18      $C[x, y] \leftarrow B[x + 2y, y]$ 
  /* ShiftRow for the second block */
19 for  $x = 0$  to 2 do
20   for  $y = 0$  to 2 do
21      $D[x, y] \leftarrow \text{RotR}(C[x, y], \rho_2(x, y))$ 
  /* AddConstant and xor state */
22  $D \leftarrow D \oplus RC[i]$ 
23  $K \leftarrow K \oplus D$ 
24 return  $K$ 
```

Algorithm 4: The Key Expansion for MOFANG-768/1024

Input: State M_1, M_2, M_3 (input length: $576 * 3$)

Output: State K

```
1  $A \leftarrow M_1$ 
  /* Transposition of words for the first block */
2 for  $x = 0$  to 2 do
3   for  $y = 0$  to 2 do
4      $B[x, y] \leftarrow A[y, 2x + y]$ 
5  $K \leftarrow B$ 
6  $A \leftarrow M_2$ 
  /* S-box for the first row for the second block */
7  $B[0] \leftarrow \text{SBox}(A[0])$ 
8 for  $x = 1$  to 2 do
9    $B[x] \leftarrow A[x]$ 
  /* Transposition of words for the second block */
10 for  $x = 0$  to 2 do
11   for  $y = 0$  to 2 do
12      $C[x, y] \leftarrow B[x + 2y, x]$ 
  /* ShiftRow for the second block */
13 for  $x = 0$  to 2 do
14   for  $y = 0$  to 2 do
15      $D[x, y] \leftarrow \text{RotR}(C[x, y], \rho_1(x, y))$ 
16  $K \leftarrow K \oplus D$ 
17  $A \leftarrow M_3$ 
  /* S-box for the second row for the third block */
18  $B[0] \leftarrow \text{SBox}(A[1])$ 
19 for  $x = 1$  to 2 do
20    $B[x] \leftarrow A[x]$ 
  /* Transposition of words for the third block */
21 for  $x = 0$  to 2 do
22   for  $y = 0$  to 2 do
23      $C[x, y] \leftarrow B[y, 2x + y]$ 
  /* ShiftRow for the third block */
24 for  $x = 0$  to 2 do
25   for  $y = 0$  to 2 do
26      $D[x, y] \leftarrow \text{RotR}(C[x, y], \rho_2(x, y))$ 
  /* AddConstant and xor state */
27  $D \leftarrow D \oplus RC[i]$ 
28  $K \leftarrow K \oplus D$ 
29 return  $K$ 
```
